DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
UNIVERSITY OF BARISHAL
FINAL EXAMINATION
Course Title: Data Structures
Course Code: CSE-1201
2nd Semester
Session: 2020-21

Time: 3 Hours **Marks:** 60

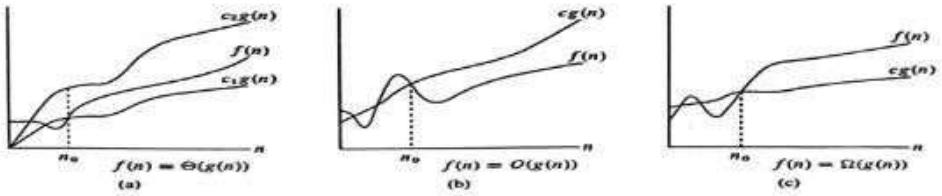
(Answer any FIVE questions)

1. a) Define data structure. List out the areas in which data structures are applied extensively. [2]

b) Define heap. Sort the items using Heap Sort algorithm: 7, 19, 19, 13, 31, 8, 18, 44, 63. [4]

c) Define priority queue. What do you mean by growth of function? Briefly describe the following (a), (b) and (c) of the figure. [3]

Figure for Q-1(c)



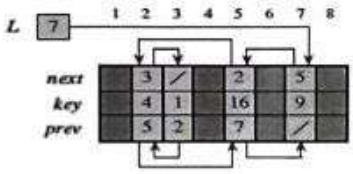
d) Represent the *tower of hanoi* problem using master theorem of recurrence. [3]

2. a) What do you mean by linked list? Implement stack's push and pop operation with linked list programmatically. [3]

b) Write down the insert and delete operation programmatically in doubly linked list. [3]

c) Why we use this type of representation instead of .....? Fill the blank with appropriate data structure name. Implement (only simulation) that data structure from the above figure. [3]

Figure for Q-2(c)



d) Define divide and conquer. Write down the pseudo code for finding the result of 6^{56} % 9 using divide and conquer. [3]

3. a) Consider the following AVL tree: [3]
i) Insert the key 30 into the tree and rebalance if needed. Draw the final and all intermediate trees that you need.
ii) Remove the key 40 from the original tree and rebalance if needed. Draw the final tree and all intermediate trees that you need.

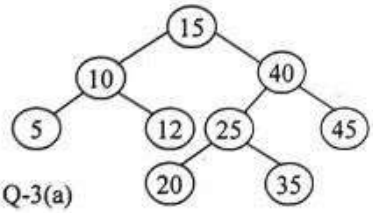


Figure for Q-3(a)

b) Write down the differences between **direct addressing table and hashing**. The figure defines collision resolution by chaining. Write down the reason of occurring collision in hash table. Write down the problems which will occurred by that method. Do you know any other method for avoiding collision? Explain. [3]

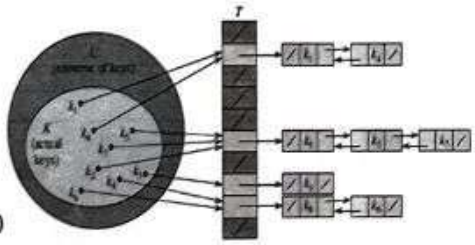


Figure for Q-3(b)

c) If start is pointing to first node of the following linked list in the following graph, then consider the following statement- [2]

start=start->next;
current=start->next;

what will be the value of address field of current?

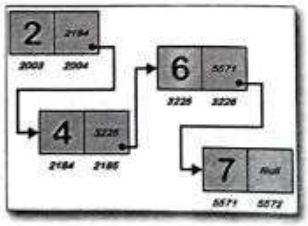


Figure for Q-3(c)

d) Write short notes on Binary search tree(BST), AVL tree and red black tree. [4]

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4. a) Consider the following sequence of stack operations: [4]

$push\ (d), push\ (h), pop\ (), push(f), push(s), pop\ (), pop\ (), push\ (m)$

i) Assume the stack is initially empty, what is the sequence of popped values, and what is the final state of the stack? (Identify which is the top of the stack)

ii) Suppose you were to replace the push and pop operations with enqueue and dequeue respectively. What would be the sequence of dequeued values, and what would be the final state of the queue? (Identify which end is the front of the queue).

b) What is meant by Underflow and Overflow on stack and queue operation? [2]

c) What is the data structure(s) used to perform recursion? Give example. [2]

d) Write a recursion algorithm to compute the nth Fibonacci number. Illustrate it drawing recursion tree. [2]

e) What are the differences between heap and BST? When to use a heap and when to use a BST? [2]

5. a) Consider the following adjacent list for directed graph. [5]

adj(y) = [x], adj(x) = [z], adj(z) = [y,w], adj(w) = [x], adj(s) = [z,w], adj(v) = [s,w], adj(t) = [u,v], adj(u) = [v].

i. Draw a graph from the above information

ii. Draw the adjacency list and adjacency matrix for the above graph. What are the advantages of adjacency matrix over adjacency list for graph traversal?

iii. Identify the tree edge, back edge, cross edge and forward edge for the graph.

b) Find out the Strongly Connected Components (SCC) algorithms using DFS for the following figure. [4]

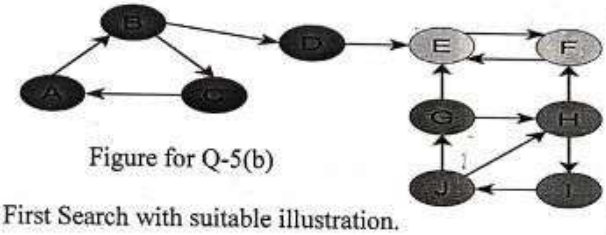

Figure for Q-5(b)

c) Write down the algorithm for the Breadth First Search with suitable illustration. [3]

6. a) Do you agree with "Kruskal algorithm as a greedy algorithm"? Justify your answer. [2]

b) What do you mean by MST? Find the MST for the following graph using PRIM'S algorithm. Is prims a Greedy algorithm? [4]

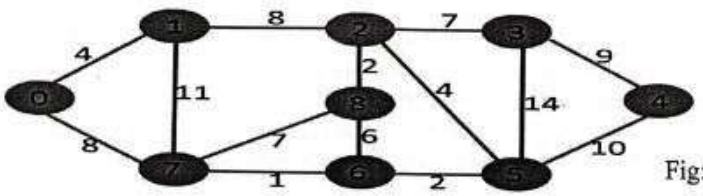

Fig: Q-6(b)

c) Define disjoint set. Write down the Union Find Algorithms along with its uses. [3]

d) **What do you mean by disjoint set data structure?** Suppose that CONNECTED-COMPONENTS is run on the undirected graph G=(V, E), where V = { a, b, c, d, e, f, g, h, i, j, k} and the edges of E are processed in the order (d, i), (f, k), (g, i), (b, g), (a, h), (i, j), (d, k), (b, j), (d, f), (g, j), (a, e). **List the vertices in each connected component after each iteration of lines 3–5.** [3]

```
CONNECTED-COMPONENTS(G)
1  for each vertex v ∈ G.V
2      MAKE-SET(v)
3  for each edge (u, v) ∈ G.E
4      if FIND-SET(u) ≠ FIND-SET(v)
5          UNION(u, v)
```

Fig: Q-6(d)

7. a) Demonstrate what happens when we insert the keys 5, 28, 19, 15, 20, 33, 12, 17, 10 into a hash table with collisions resolved by chaining. Let the table have 9 slots, and let the hash function be h(k) = k mod 9. [3]

b) Consider a hash table that uses the linear probing technique with the following hash function f(x)=(5x+4) %11. (The hash table is of size 11.) If you insert the values 3,9,2,1,14,6, and 25 into the table, in that order, show where these values would end up in the tables? [3]

index	0	1	3	4	5	6	7	8	9	10
value										

c) Traverse the given tree using In-order, Preorder and Post-order traversals. [2]

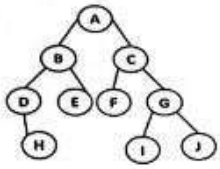

Fig: Q-7(c)

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d) Consider a hash table of size 7 with hash function $h(k) = k \mod 7$. Draw the table that results after inserting in the given order, the values, 19, 26, 13, 48, 17 for each of the scenarios below: [4]

i) When collisions are handled by separate chaining.

ii) When collisions are handled by linear probing.

iii) When collisions are handled by double hashing.

iv) Using a second hash function $h'(k) = 5 - (k \mod 5)$.

8. a) Why quick sort is also called as partition and exchange sort? Derive its best- and worst-case running time. [4]

b) Illustrate the operation of Insertion sort on the array A = < 3, 2, 6, 4, 1 >. Analyze the best and worst case analysis for the insertion sort algorithm. [3]

c) Algorithms A and B spend exactly $T_A(n) = c_A n \log_2 n$ and $T_B(n) = c_B n \log_2 n$ microseconds, respectively, for a problem of size n. Find the best algorithm for processing n = 2^20 data items if the algorithm A spends 10 microseconds to process 1024 items and the algorithm B spends only 1 microsecond to process 1024 items. [2]

d) For the set of {1, 4, 5, 10, 16, 17, 21} of keys, draw binary search trees of heights 4, 5, and 6. Define predecessor and successor along with the uses. [3]

Good Luck!!!

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